# AIGuessr — User Guide

A GeoGuessr-inspired serious game to discover the material footprint of AI.

This guide explains how to set up and run an AIGuessr session, whether for a full jigsaw classroom workshop (90 min) or a quick game-only session (25 min).

## Requirements

- A computer with a projector (presenter view)
- Students with smartphones (player view)
- An internet connection
- No installation required: the app runs in any web browser

## Getting Started

### Option A: Use the hosted instance

Open the hosted instance in your browser (address withheld for review). No setup needed.

### Option B: Run locally

Clone the repository, install dependencies, and start:

```
git clone <repository URL>.git && cd aiguessr && npm install && npm start
```

(see README)

The app will be available at http://localhost:5173 (frontend) and http://localhost:3001 (backend).

## Configuration Screen

The configuration screen lets you set up the game session:

| Setting | Description | Recommended |
|---|---|---|
| Number of teams | How many teams compete | 4-6 teams |
| Timer duration | Seconds per question | 60-90 seconds |
| Number of questions | Questions to play | 5 (short) to 20 (full) |
| Multiplayer mode | Students join via QR code | Always ON |
| Questionnaire mode | Pre/post Likert survey | ON for research |
| Category filter | Select specific categories | All categories |

## Game Modes

### Full Workshop (90 min) — Jigsaw Classroom

| Phase | Duration | Description |
|---|---|---|
| Team formation | 5 min | Form teams of 4-5 students |
| Expert sheets | 15 min | Each member studies one of the 4 expert sheets |

| | | |
|---|---|---|
| Connection | 5 min | Students scan QR code, demo question |
| Game session | 40 min | 18-20 questions across all 4 categories |
| Debriefing | 15 min | Discussion of key discoveries |
| Questionnaire | 5 min | Post-game survey (12 items + open) |

### Quick Session (25 min) — Game Only

Pre-questionnaire (3 min) then game (20 min, 5 questions, 1 per category) then post-questionnaire (3 min). No expert sheets or jigsaw structure.

## Game Flow

Each question follows this sequence:

1. Question Intro: a 3D globe zooms to the region of interest
2. Research: teams discuss and research (timer with pause option)
3. Answer: each team places a marker on the interactive map
4. Reveal: correct location shown, distances calculated, points awarded (0-1000)
5. Impact: key environmental/social data about the location

## Question Categories

| Category | Questions | Topics |
|---|---|---|
| Extraction (red) | 10 | Cobalt, lithium, rare earths, copper, tantalum, nickel, silicon, gold |
| Production (orange) | 4 | TSMC semiconductors, NVIDIA GPUs, Foxconn assembly |
| Data Centers (blue) | 5 | Virginia, Oregon, Ireland, Iowa, Sweden |
| E-Waste (green) | 2 | Agbogbloshie (Ghana), Guiyu (China) |

## Expert Sheets

Four PDF expert sheets are provided in the docs/ folder, each covering one phase of the AI lifecycle. In jigsaw mode, each team member reads a different sheet before the game begins, becoming the "expert" on that phase for their team.

## After the Game

At the end of the session, the app provides:

- Final podium with team rankings
- Per-category performance breakdown

- Overview map showing all answers across all questions
- PDF and JSON export of results (downloadable or cloud-uploaded)

## Hints & Tips

- Use the pause button during the research phase if students need more time
- The hint system provides clues at a -20% or -40% score penalty
- Dark mode is available for better projection in dimly lit rooms
- The water consumption counter (visible during data center questions) is a powerful discussion starter

## License

AIGuessr is open-source under the MIT License. You are free to use, modify, and redistribute it.

Repository: <repository URL> (see README)